Internetworking with TCP/IP

Lecture 2: Internet Protocol IPv4

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Outline

- Introduction
- Internetworking Issues
- IPv4
- Service Model
- IPv4 Datagram Format
- IP Service Primitives
- IPv4 Addressing
- Classless Addressing
- Subnetting and Supernetting
- Datagram Delivery
- Finding the Local Router
- Routers
- ICMPv4
- Address Translation
- Datagram Delivery

Introduction

- Definition **Internetworking**
  - Interconnection of multiple networks into an Internet

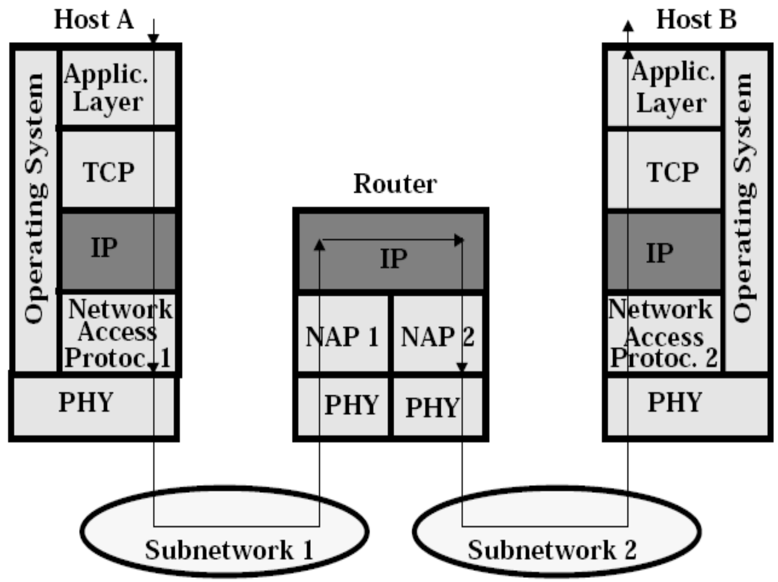
- Task: getting packets from the source to destination

- In order to achieve the goal, one needs to
  - Have knowledge about the topology of the network and also about the network itself
  - Be able to select appropriate path through the network
  - Select routes to avoid congestion
  - Be able provide internetworking when the source and destination are in different networks

- **Some definitions**
  - **Data Terminal Equipment (DTE):** End System (ES) or Host
  - **Data Communication System (DCS)**
  - **Internetwork (or Internet):** composite network being used (e.g., LAN/WAN/LAN)
  - **Subnetwork:** constituent network of the Internet (e.g., LAN)
  - **Intermediate System (IS) or Interworking Unit (IWU):** device that interconnects two networks (e.g., bridge, router, gateway)

Interworking Issues

- **Functions**
  - **TCP:** e2e error and flow control
  - **IP:** routing and forwarding
  - **NAP:** forwarding (only within subnetwork); point-to-point error and flow control;



Internetworking Issues (cont)

- **Some of the most important issues**
  - Overall requirements
  - Services provided to transport layer
  - Naming
  - Addressing
  - Routing
  - Internetworking
  - Quality of Service (eventually)
  - Maximum packet size
  - Flow and error control (at DLL and TL only; not at IP)
  - Congestion control
  - Error reporting
  - Network layer structure
  - Architectural approaches
  - …

Interworking Issues (cont)

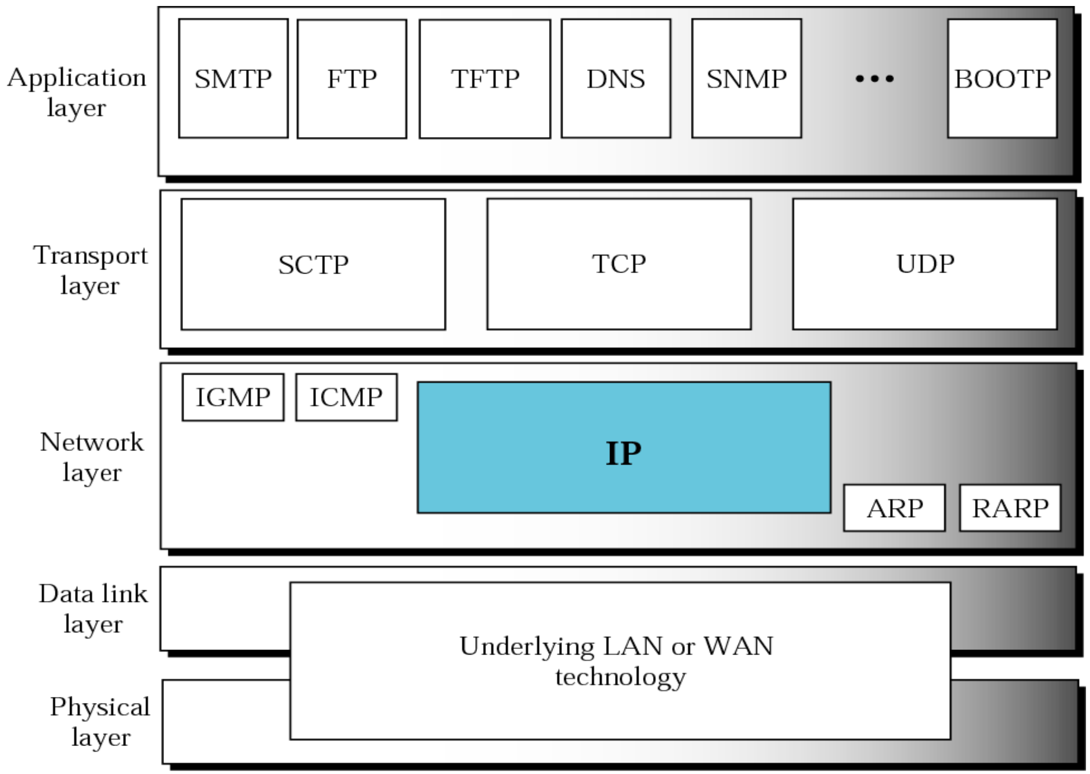
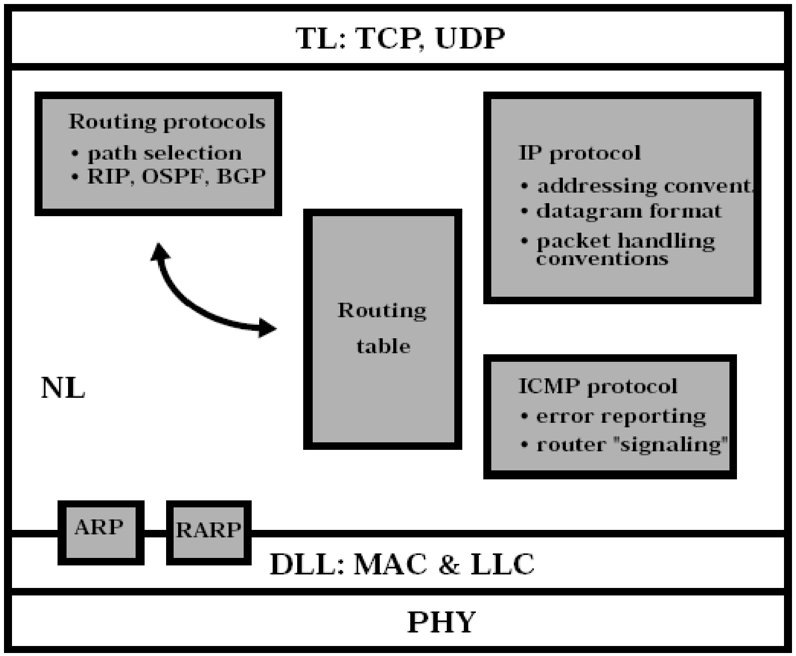
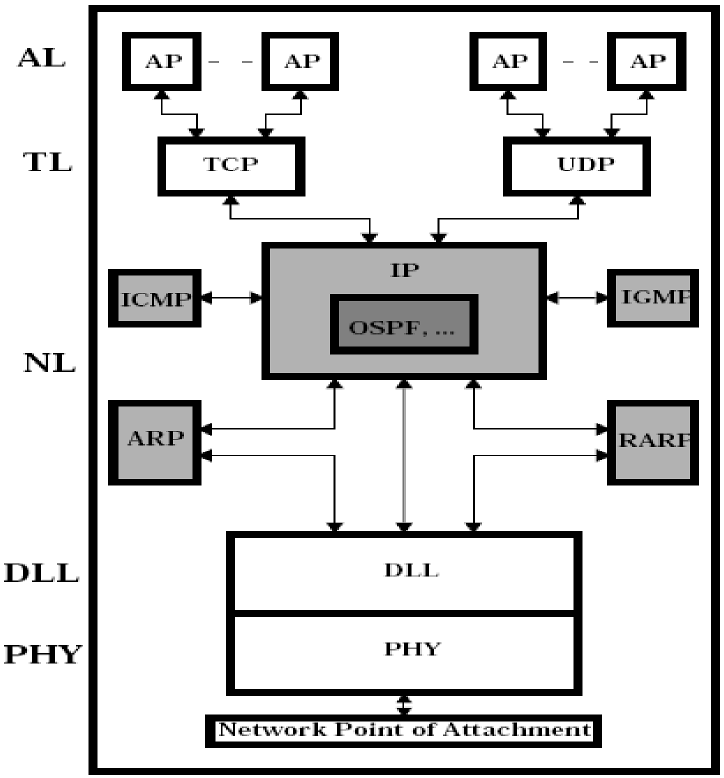


Fig 7.1 – Forouzan 4th edition

Internetworking Issues (cont)



Internet network layer



IP adjunct protocols

IPv4

- **Two main components**
  - IP services (IP to/from TCP/UDP)
  - Protocol format and mechanisms

- **Aspects considered**
  - Service model
  - IPv4 datagram format
  - IP service primitives
  - IPv4 addressing
  - Subnets and subnet masks
  - Special purpose addresses
  - Segmentation and reassembly
  - ICMPv4
  - Address translation
  - Finding the local router
  - Datagram delivery
  - Generic router architecture

- Important addresses
  - RFC editor homepage http://www.rfc-editor.org
  - RFC hypertext archive http://rfc.dotsrc.org/
  - InterNIC (Internet Domain Name Registration Services) http://www.internic.net/
  - IANA (Internet Assigned Numbers Authority) http://www.iana.org
  - ICANN (Internet Cooperation for Assigned Names and Numbers) http://www.icann.org

Service Model

- **IP service model**
  - Best-effort, connectionless service, i.e., no guarantees, only datagram delivery

- **Main components**
  - Addressing scheme
  - Datagram model of data delivery

- **Main IP Specs**
  - IPv4: RFC 791 (1981)
  - ICMP: RFC 792 (1981)
  - ARP (Ethernet): RFC 826, 1027 (1982)
  - RARP (Ethernet): RFC 903 (1982)

- **ARP/RARP**
  - There are more versions ARP/RARP, depending upon the network below
  - ARP cache: contain address mappings IP address vs. PHY address

IPv4 Datagram Format

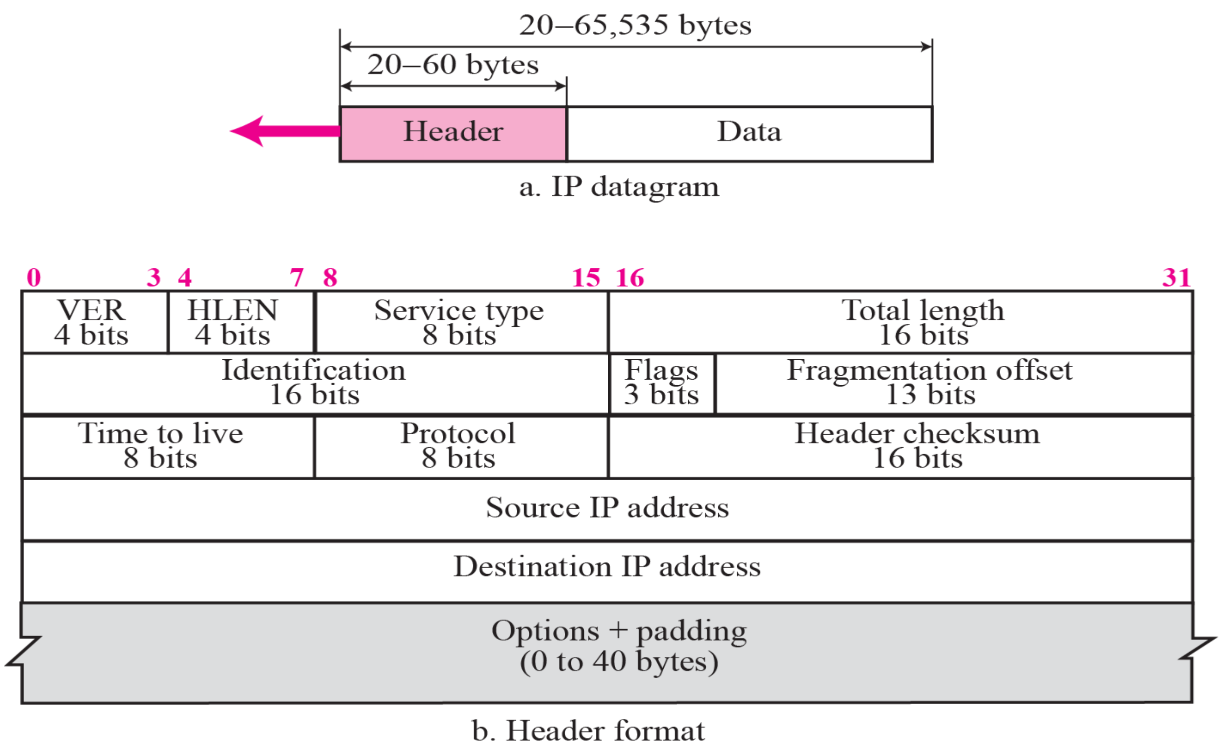


Fig 7.2 – Forouzan 4th edition

IPv4 Datagram Format (cont)

- **Version (4 bits):** indicates protocol version number (IPv4; IPv6)
- **Header Length (HLEN – 4 bits):** length of header in 32-bit words
- **Differentiated Services (DS),** previously called **Service Type (ST)**
    - **DS:** first 6 bits make up the code point subfield, and the last 2 bits are not used
    - **ST:** first 3 bits are called “Precedence” bits, the next 4 bits are called “Type of Service (ToS)” and the last bit is not used
- **Total Length (16 bits):** total datagram length, including the header (max 65,535 bytes)
- **Identification (16 bits):** used to manage the segmentation/reassembly process
- **Flags (3 bits):** only two bits (“More” and “Don’t Fragment”) are currently used to manage the segmentation/reassembly process
- **Fragment Offset (13 bits):** indicates position where in the original datagram the specific fragment belongs; measured in 64-bit units
- **Time to Live (8 bits):** TTL specifies how long (in sec), a datagram is allowed to remain in the Internet; decreased each time the datagram passes a router (1 sec/router)
- **Protocol (8 bits):** indicates the higher layer protocol to received the data field (at destination) – RFC 1700
- **Header Checksum (16 bits):** an error-detecting code is applied in the IP header only; this code is verified and recomputed at every router
- **Source Address (32 bits):** to allow a variable number of bits to specify the network (and the subnetwork) and the host address of the transmitting host
- **Destination address (32 bits):** to allow a variable number of bits to specify the network (and the subnetwork) and the host address of the receiving host
- **Options (variable):** rarely used today, it allows an IP header to be extended with diverse options, e.g., security, source routing, route recording, stream identification; time stamping
- **Padding (variable):** used to ensure that the datagram header has a length that is multiple of 32 bits
- **Data (variable):** must have a length that is an integer multiple of 8 bits; maximum length is 65,536 octets

IPv4 Datagram Format (cont)

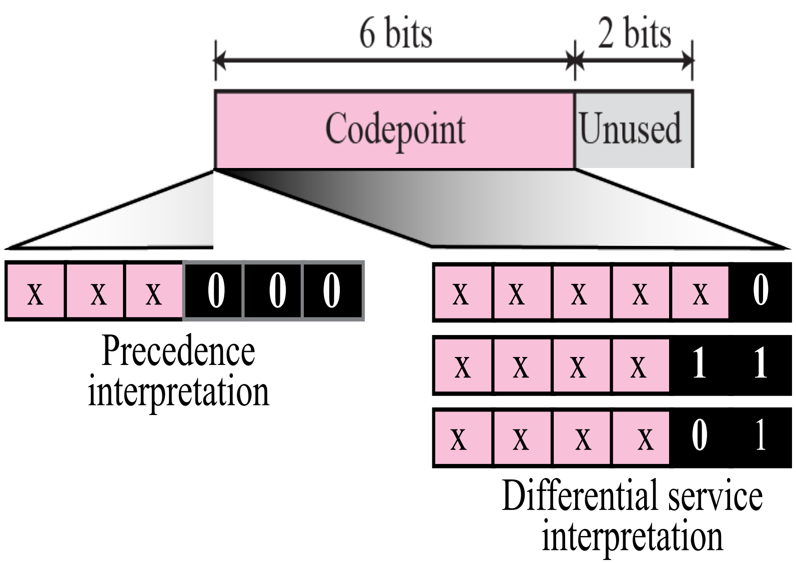


Fig 7.3 – Forouzan 4th edition

TOS Bits	*Description*
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay

Table 8.1 – Forouzan 3rd edition

Protocol	*TOS Bits*	*Description*
ICMP	0000	Normal
BOOTP	0000	Normal
NNTP	0001	Minimize cost
IGP	0010	Maximize reliability
SNMP	0010	Maximize reliability
TELNET	1000	Minimize delay
FTP (data)	0100	Maximize throughput
FTP (control)	1000	Minimize delay
TFTP	1000	Minimize delay
SMTP (command)	1000	Minimize delay
SMTP (data)	0100	Maximize throughput
DNS (UDP query)	1000	Minimize delay
DNS (TCP query)	0000	Normal
DNS (zone)	0100	Maximize throughput

Table 8.2 – Forouzan 3rd edition

Service Type

IPv4 Datagram Format (cont)

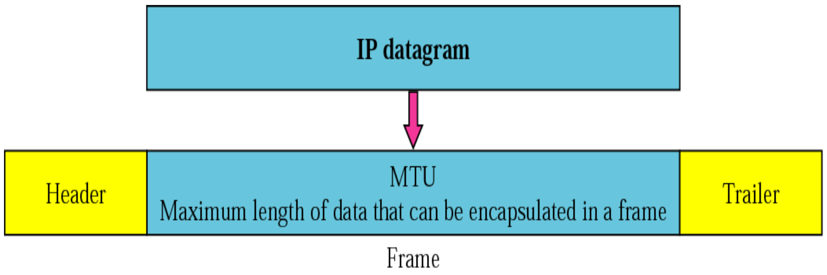


Fig 7.6 – Forouzan 4th edition

Protocol	*MTU*
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

Table 8.5 – Forouzan 3rd edition

Datagram Maximum Transfer Unit

IPv4 Datagram Format (cont)

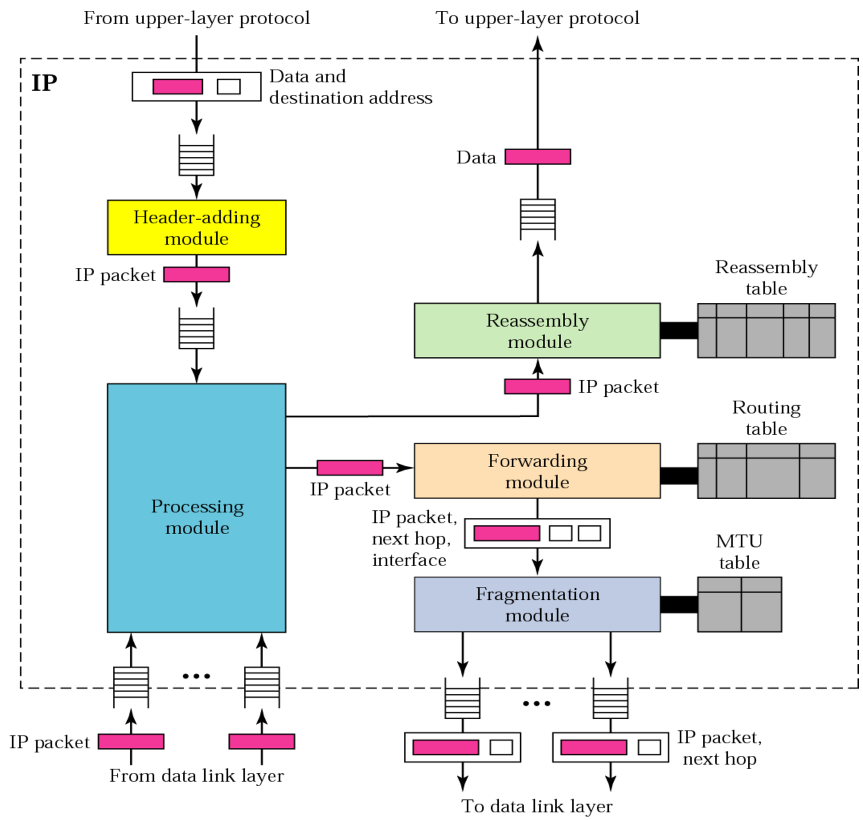


Fig 7.29 – Forouzan 4th edition

IP components

IP Service Primitives

TCP/UDP --> IP	IP --> TCP/UDP
Send {	Deliver {
Source address	Source address
Destination address	Destination address
Protocol	Protocol
Type of service	Type of service
Identification	---
Don't fragment identifier	---
TTL	---
Data length	Data length
Options data	Options data
Data	Data
}	}

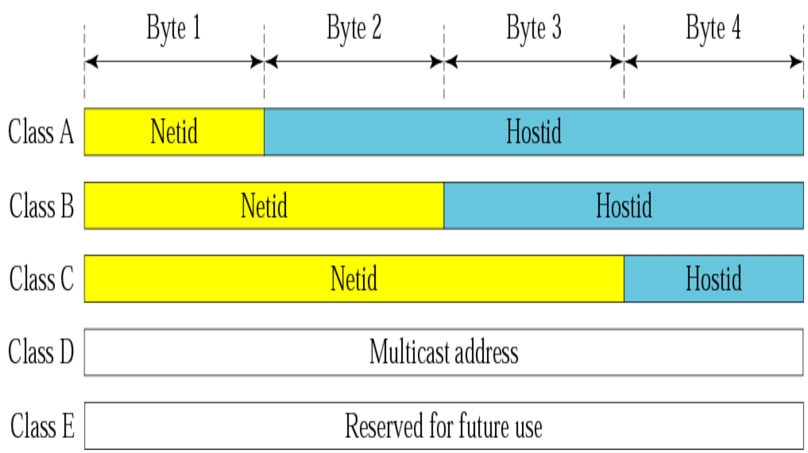
IPv4 Addressing

- An IP address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet
- IP addresses are unique in the sense that each address defines one, and only one, connection to the Internet
- The address space of IPv4 is 2^{32} or 4,294,967,296
- Two categories of IP addressing
  - Classful addressing
  - Classless addressing

IPv4 Addressing (cont)

Address format: Prefix/Network/Host

Dotted decimal representation: xxx.yyy.zzz.vvv (x, y, z and v between 0 and 9)



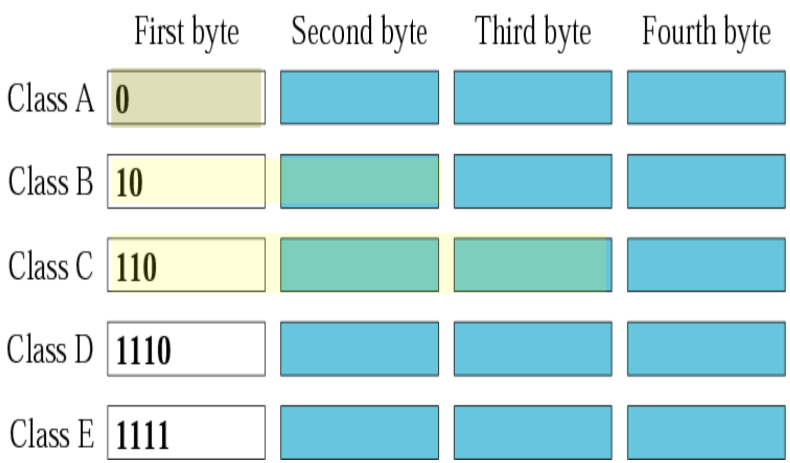
Netid and hostid

Fig 5.8 – Forouzan 4th edition

- **Class A:** 1.0.0.0 to 127.255.255.255 (7 bits for Netid and 24 bits for Hostid; 126 network addresses: 1 – 126; 0 and 127 are reserved; max 16,777,214 hosts/net)
- **Class B:** 128.0.0.0 to 191.255.255.255 (14 bits for Netid and 16 bits for Hostid; 2^{14} = 16,384 network addresses; max 65,534 hosts/net)
- **Class C:** 192.0.0.0 to 223.255.255.255 (21 bits for Netid and 8 bits for Hostid; 221 = 2,097,152 network addresses; max 255 hosts/net)
- **Class D:** 224.0.0.0 to 239.255.255.255 (228 network addresses) [RFC 1112]
- **Class E:** 240.0.0.0 to 247.255.255.255

- **Special addresses**
  - 0.0.0.0 – some unknown host (booting purposes)
  - 10.0.0.1 – ARPA NIC (old)
  - 127.0.0.0 – this host (local loop)
  - 255.255.255.255 – any host (broadcast purposes)
  - A.255.255.255 – direct broadcast (class A)
  - B.B.255.255 – direct broadcast (class B)
  - C.C.C.255 – direct broadcast (class C)

IPv4 Addressing (cont)



Finding the class in binary notation

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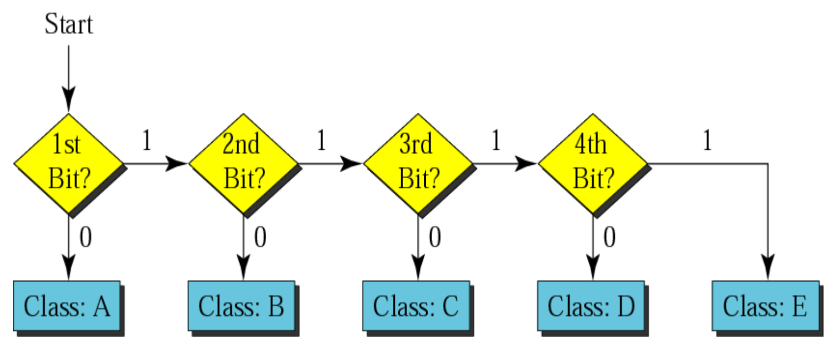

Finding the address class

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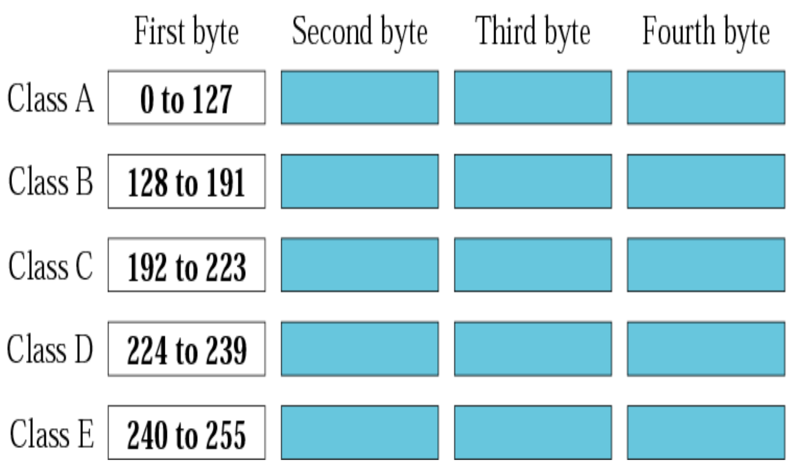
IPv4 Addressing (cont)

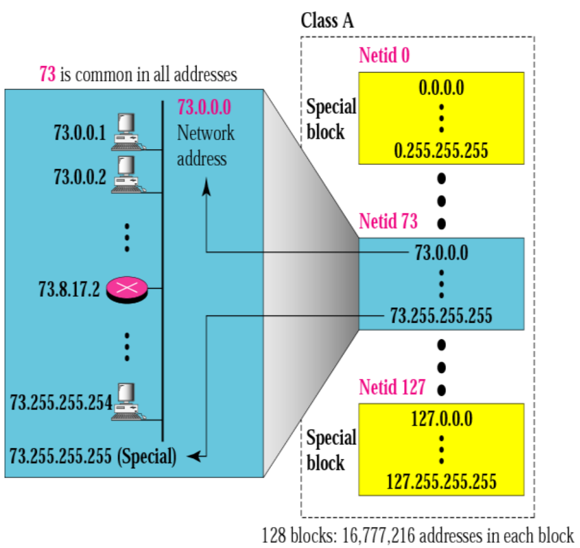


Finding the class in decimal notation

Fig 5.6 – Forouzan 4th edition

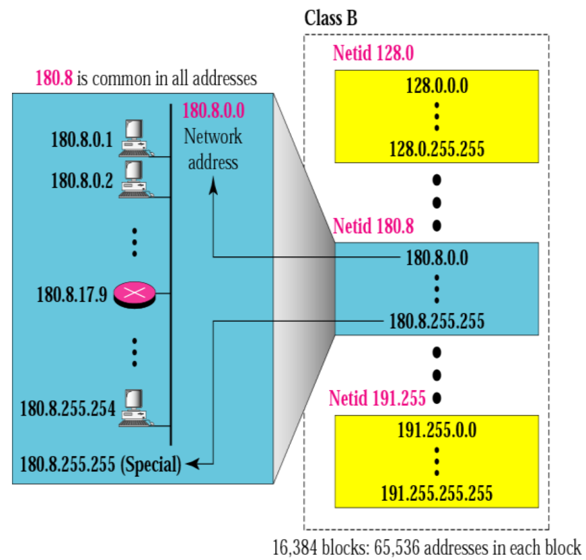
- The network address is assigned to organization
- The range of addresses can automatically be inferred from the network address

IPv4 Addressing (cont)



Blocks in class A

Fig 4.7 – Forouzan 3rd edition



Blocks in class B

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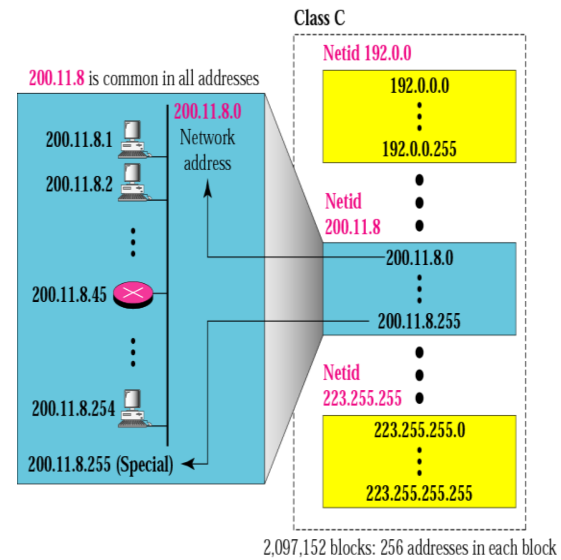


Blocks in class C

Fig 4.9 – Forouzan 3rd edition

IPv4 Addressing (cont)

- Multicast addresses

Address	*Group*
224.0.0.0	Reserved
224.0.0.1	All SYSTEMS on this SUBNET
224.0.0.2	All ROUTERS on this SUBNET
224.0.0.4	DVMRP ROUTERS
224.0.0.5	OSPFIGP All ROUTERS
224.0.0.6	OSPFIGP Designated ROUTERS
224.0.0.7	ST Routers
224.0.0.8	ST Hosts
224.0.0.9	RIP2 Routers
224.0.0.10	IGRP Routers
224.0.0.11	Mobile-Agents

Category addresses

Table 4.5 – Forouzan 3rd edition

Address	*Group*
224.0.1.7	AUDIONEWS
224.0.1.10	IETF-1-LOW-AUDIO
224.0.1.11	IETF-1-AUDIO
224.0.1.12	IETF-1-VIDEO
224.0.1.13	IETF-2-LOW-AUDIO
224.0.1.14	IETF-2-AUDIO
224.0.1.15	IETF-2-VIDEO
224.0.1.16	MUSIC-SERVICE
224.0.1.17	SEANET-TELEMETRY
224.0.1.18	SEANET-IMAGE

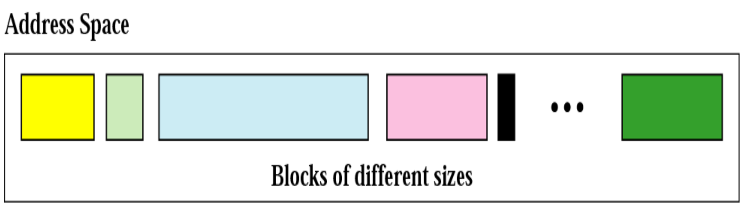
Addresses for conferencing

Table 4.6 – Forouzan 3rd edition

Classless Addressing

- Big problem related to IP addressing: **ROADS (Running Out of Address Space)**
- Examples: exhaustion of class B address space and explosion of routing table growth triggered by a flood of new class C
- Solution advanced by IETF: **Classless Inter-Domain Routing (CIDR)**
- Today, we have therefore two forms of IP addressing
  - Classfull addressing: N (Prefix and Netid) is constrained to 8 or 16 or 24 bits only [RFC 1700 and RFC 1117]
  - Classless addressing: N can be any number of bits, up to 30 [RFC 1518 and RFC 1519]; used in CIDR
- Accordingly, CIDR dotted decimal representation is in the form **xxx.yyy.zzz.vvv/ww**, where **ww** indicates the number of leading bits in the 32-bit IP address that constitute the "Network" portion (example 223.1.1.0/24)
- Classful addressing is a special case of classless addressing

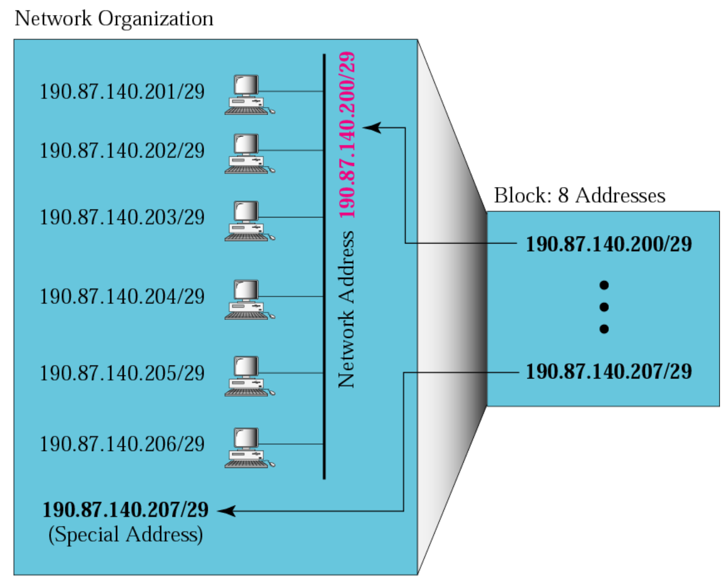
Classless Addressing (cont)



/n	Mask	/n	Mask	/n	Mask	/n	Mask
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

Prefix lengths

Table 5.1 – Forouzan 3rd edition



Example

Fig 5.5 – Forouzan 3rd edition

Subnetting and Supernetting

- A third hierarchical level was introduced in 1984 in the addressing structure, namely **subnet**
- The address format is therefore: **Prefix/Network/Subnetwork/Host**
- The Subnet field can be of any length and it is specified by a **32-bit mask**
- The IP addresses are therefore designed with two levels of hierarchy
- One determines that an address belongs to a subnet by a ”**comparison-under-mask**” operation (bitwise AND)
- The effect of the Subnet Mask is to erase the portion of the *Host* field, so leaving only the **Network** number and the **Subnetwork** number

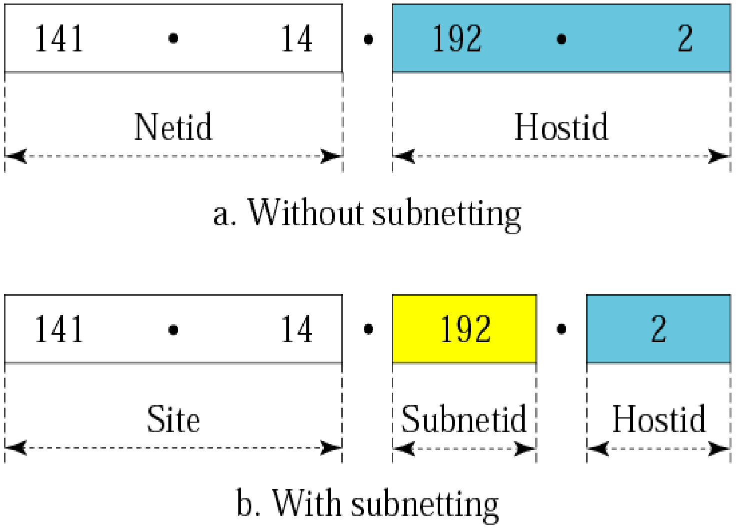
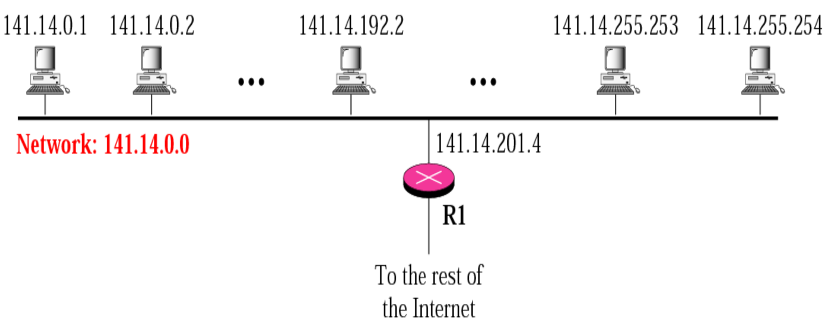


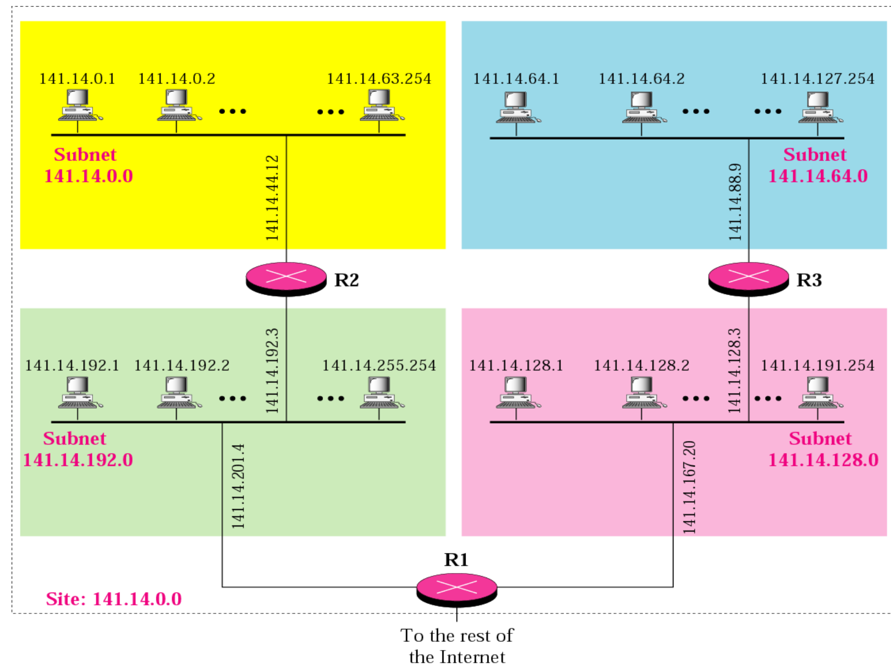
Fig 4.22 – Forouzan 3rd edition

Subnetting and Suppernetting (cont)



A network with two levels of hierarchy (not subnetted)

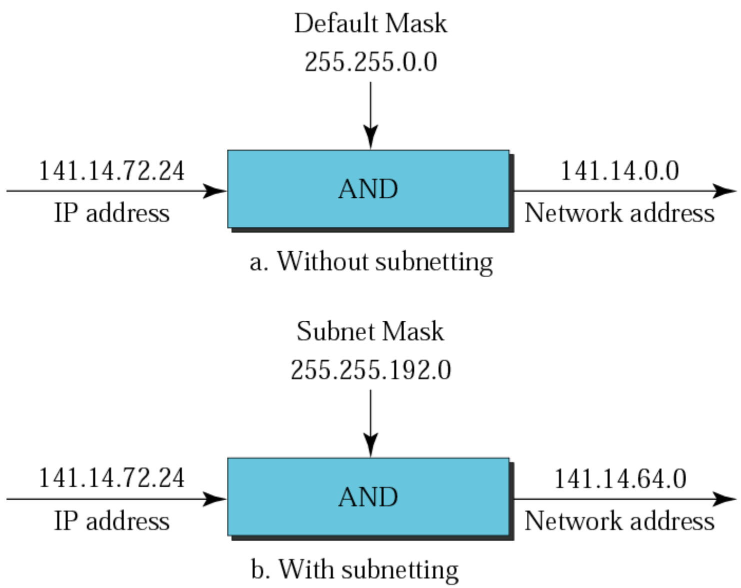
Fig 4.20 – Forouzan 3rd edition



A network with three levels of hierarchy (subnetted)

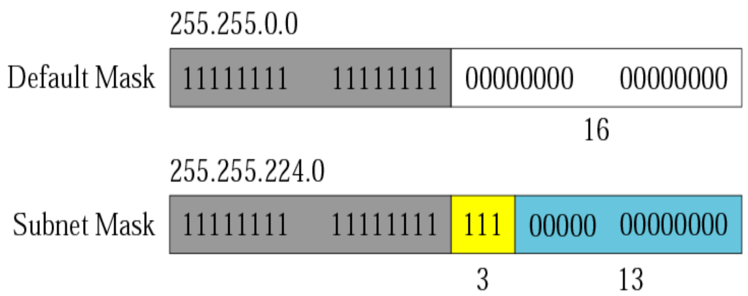
Fig 4.21 – Forouzan 3rd edition

Subnetting and Suppernetting (cont)



Default mask and subnet mask

Fig 4.24 – Forouzan 3rd edition

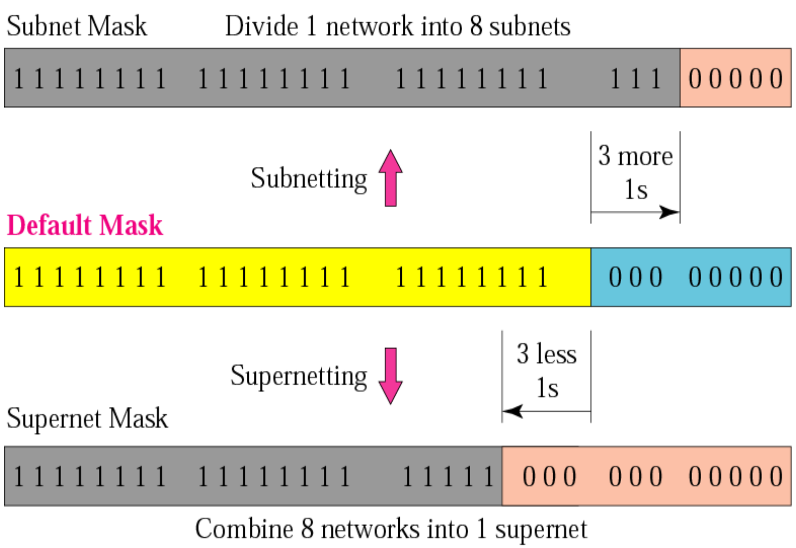


Comparison of a default mask and a subnet mask

Fig 4.25 – Forouzan 3rd edition

Subnetting and Suppernetting (cont)

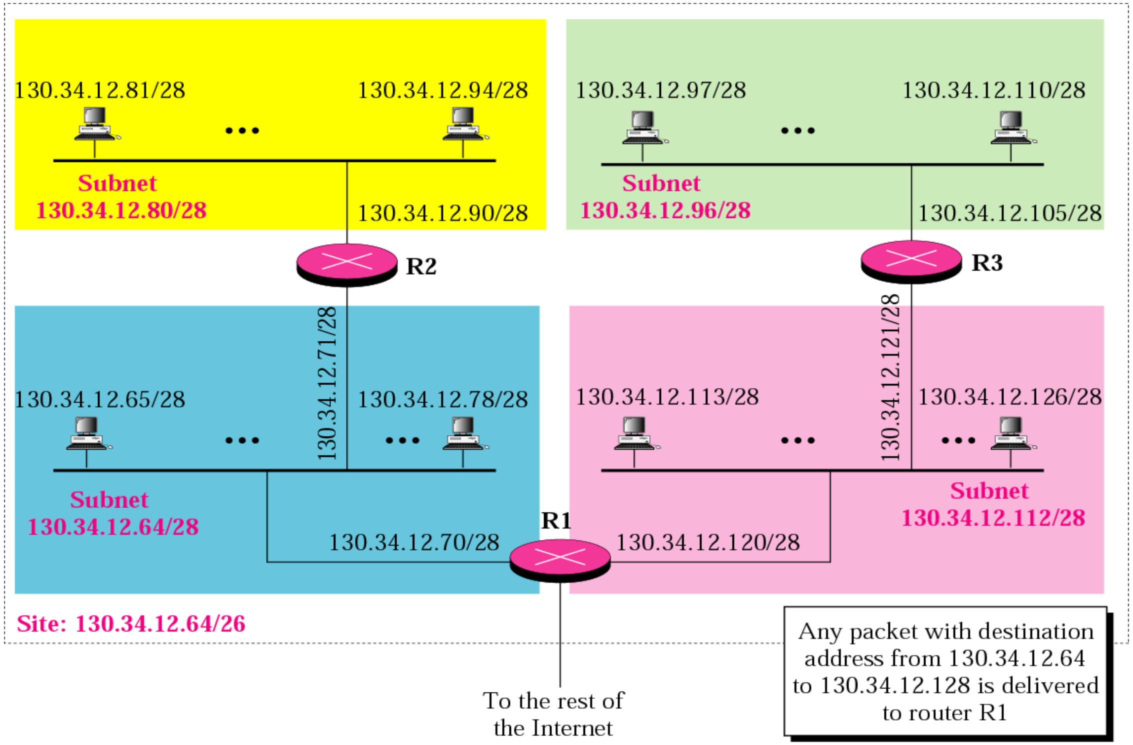
- In subnetting, one needs the first address of the subnet and the subnet mask to define the range of addresses
- In supernetting, one needs the first address of the supernet and the supernet mask to define the range of addresses
- The idea of subnetting and supernetting of classful addresses is almost obsolete



Comparison of subnet, default and supernet masks

Fig 4.27 – Forouzan 3rd edition

Subnetting and Suppernetting (cont)



Example of subnetting

Fig 5.6 – Forouzan 3rd edition

Datagram Delivery

IP Routing Algorithm (for both hosts and routers):

Route_IP_Datagram (Datagram DAT, Routing_Table RT, Subnet_Mask M)

extract Destination IP address (D) from the Datagram DAT
and compute the Network Prefix (N) by bitwise-AND of D and Subnet
Mask M

if N matches any directly connected network address (own interfaces)

  deliver datagram to destination D over that network
(this involves resolving D to a physical address, encapsulating the
Datagram DAT and sending the frame)

else if RT contains a host-specific route for D

  send DAT to the next-hop router specified in RT
  (obs: next-hop MUST lie on a directly connected network)

else if RT contains a route for network N

  send DAT to the next-hop router specified in RT

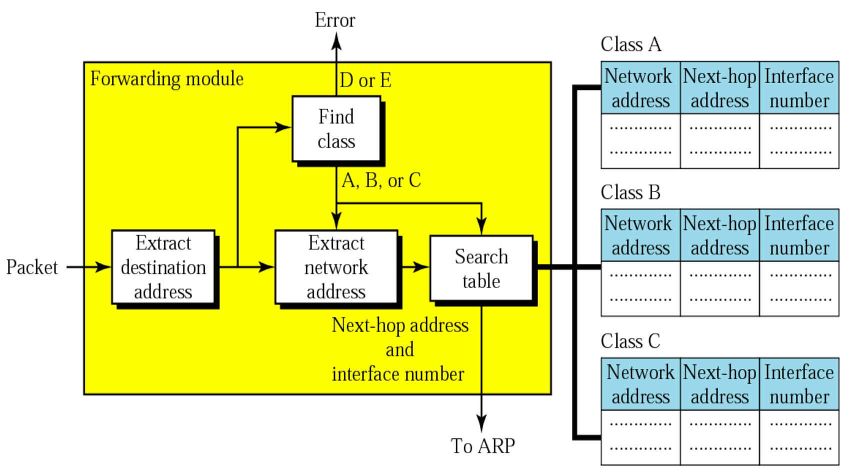
else if RT contains a default route for network N

  send DAT to the default router specified in RT

else if no matches are found

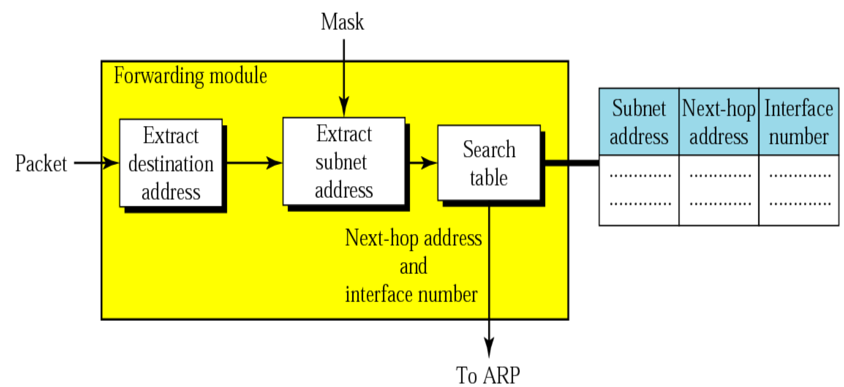
  declare a routing error;

Datagram Delivery (cont)



Simplified module for forwarding in classful address without subnetting

Fig 6.7 – Forouzan 4th edition



Simplified module for forwarding in classful address with subnetting

Fig 6.10 – Forouzan 4th edition

Datagram Delivery (cont)

- In classful addressing, one can have a routing table with three columns
- In classless addressing, one needs at least four columns

Mask	Network address	Next-hop address	Interface	Flags	Reference count	Use
....	....	....	....	....	....	....

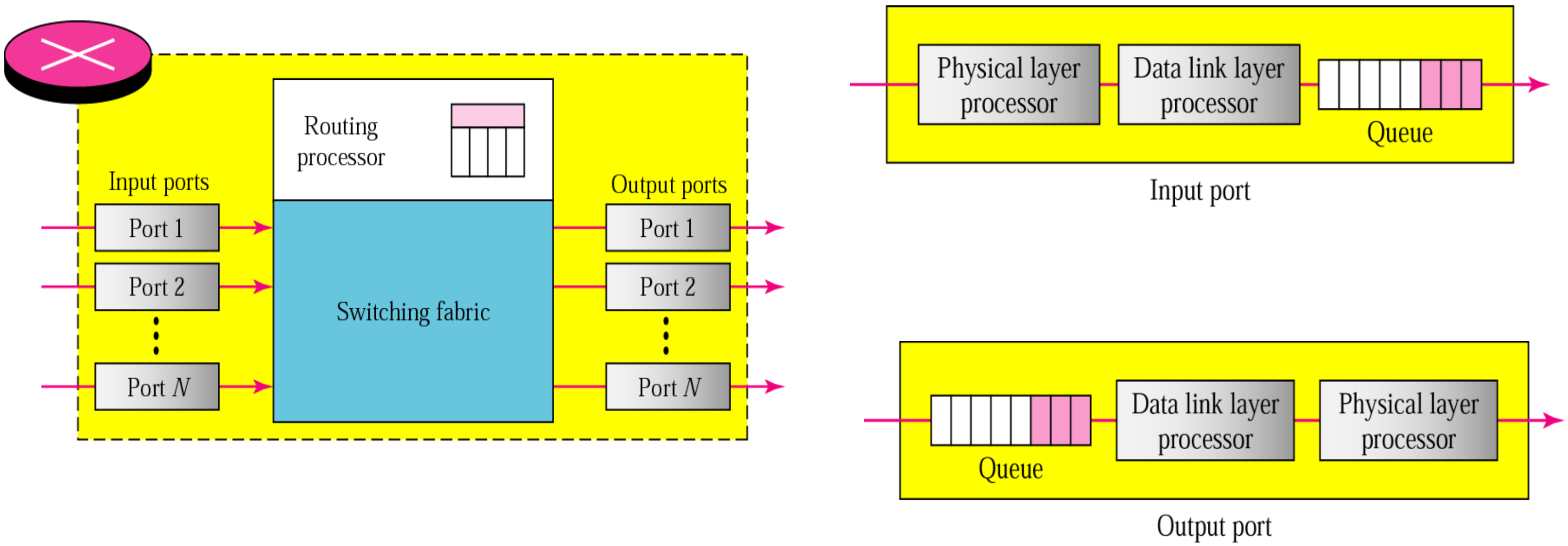
Common fields in a routing table

Fig 6.18 – Forouzan 3rd edition

Finding the Local Router

- Question: what to do when there are more routers in a network, and not one? To which router send the datagram?
- Another question: how to learn the addresses of these routers?
- **Discovery procedures** are used to answer these questions
- **Discovery procedures**
  - **Static** (system manager, difficult)
  - **Dynamic**, by using ICMP messages of type **advertisements** and **solicitations**
- **Advertisements** are broadcasted by the routers to the Internet (every 7th minute)
- **Solicitations:** the hosts may trigger the emission of these messages

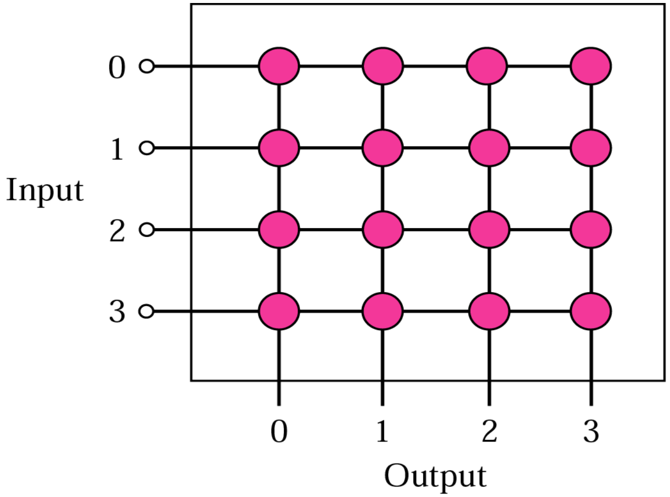
Routers



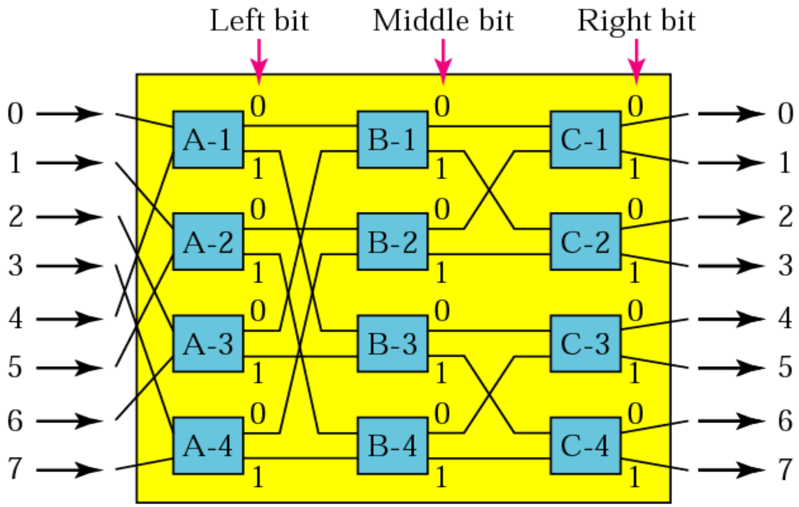
Router components

Figs 6.22, 6.23 and 6.24 – Forouzan 4th edition

Switches



Crossbar switch

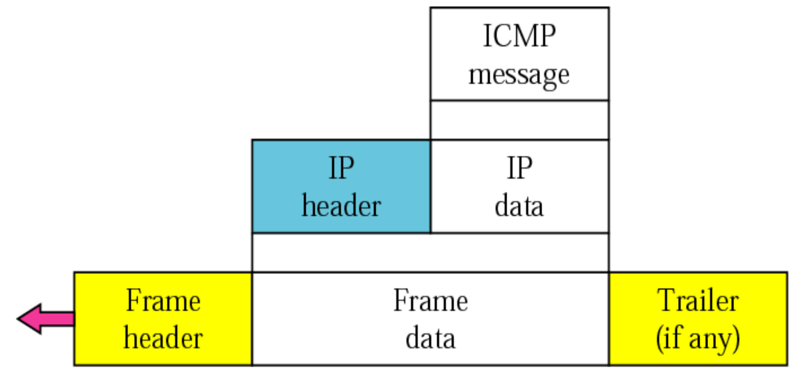


Banyan switch

Figs 6.25 and 6.26 – Forouzan 4th edition

ICMPv4

- Used by hosts, routers and gateways to communicate information to each other, mostly for signaling/diagnostic error conditions; it provides feedback information about network problems [RFC 792]
- Most ICMP messages are diagnostic information that is sent back to the source when a router destroys a packet, e.g., when the routing can not be solved (*destination unreachable*) or when TTL expires (*time exceeded*) or when a router detects a congestion (*source quench*)
- ICMP also defines an *echo* function that can be used for testing connectivity (*ping*)
- ICMP is considered as part of IP but architecturally it lies just above IP, as ICMP messages are carried inside IP datagrams
- Software that are using ICMP: *ping*, *traceroute*



ICMP encapsulation

Fig 9.2 – Forouzan 4th edition

ICMPv4 (cont)

- Two categories of ICMP message
  - Error-reporting messages: to report problems that a router or destination host may encounter
  - Query messages: to get specific information from a router or another host

Category	*Type*	*Message*
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply
	17 or 18	Address mask request or reply
	10 or 9	Router solicitation or advertisement

ICMP messages

Table 9.1 – Forouzan 4th edition

ICMPv4 (cont)

- ICMP messages have the format Type/Code/Checksum and also contain the first eight bytes of the IP datagram (together with the Header) that caused the ICMP message to be generated in the first place, such as the sender can determine the packet that caused the error

0	8	16	31
Type	Code	Checksum	
Unused			
IP Header	+ first 8 bytes of the original data Datagram		

ICMP Type	Code	Description several messages
0	0	echo reply (to ping)
3	0	destination network unreachable
3	1	destination host unreachable
3	2	destination protocol unreachable
3	3	destination port unreachable
3	6	destination network unknown
3	7	destination host unknown
4	0	source quench (congestion control)
8	0	echo request
9	0	router advertisement
10	0	router discovery/solicitation
11	0	TTL expired
12	0	parameter problem/IP header bad

Address Translation

- Translation between IP and MAC address
  - Address Resolution Protocol (ARP) IP to MAC [RFC 826]
  - Reverse Address Resolution Protocol (RARP) MAC to IP [RFC 903]

- ARP lies architecturally bellow IP, as ARP/RARP messages are carried inside MAC frame (as payload)

- Variants of ARP/RARP have been defined for different networking technologies, e.g., Ethernet/IEEE 802.3, Token Ring/IEEE 802.5, FDDI, etc

- Programs that use ARP: **arp** (comment: this is an Ethernet broadcast)

- Lists of IP/MAC are in Proxy ARP memory (routers/gateways) and ARP cache memory (hosts)

References

- Forouzan B.F,"TCP/IP Protocol Suite", 4th edition, McGraw-Hill, chapters 4, 5, 6, 7, 8, 9, 18
- Other useful addresses

THANK YOU!